

Predict the Output

Conditional operator and switch statement practice for Lecture 07.01

For each example, predict exactly what the program prints before running it. Pay attention to which branch executes, whether a case matches, and where break appears.

Instructor use: ask students to explain the path, not only name the output.

1. Simple conditional choice

Output: adult

Reason: The condition is true because 20 is at least 18.

```
#include <iostream>
using namespace std;

int main() {
    int age = 20;
    string label = age >= 18 ? "adult" : "minor";

    cout << label << endl;
}
```

2. Numeric conditional expression

Output: 0

Reason: The true value, 0.0, is selected because there are at least 3 items.

```
#include <iostream>
using namespace std;

int main() {
    int items = 3;
    double shipping = items >= 3 ? 0.0 : 5.99;

    cout << shipping << endl;
}
```

3. Conditional operator inside output

Output: Regular price

Reason: The condition is false, so the second string is printed.

```
#include <iostream>
using namespace std;

int main() {
    bool member = false;

    cout << (member ? "Discount applied" : "Regular price") << endl;
}
```

4. Nested conditional operator

Output: B

Reason: 85 fails the 90 test but passes the 80 test.

```
#include <iostream>
using namespace std;

int main() {
    int score = 85;

    string grade =
        score >= 90 ? "A" :
        score >= 80 ? "B" :
        score >= 70 ? "C" : "D";

    cout << grade << endl;
}
```

5. switch with a matching case

Output: Express

Reason: The value 'E' selects the second case, which ends with break.

```
#include <iostream>
using namespace std;

int main() {
    char code = 'E';

    switch (code) {
        case 'S':
            cout << "Standard";
            break;
        case 'E':
            cout << "Express";
            break;
        case 'P':
            cout << "Pickup";
            break;
        default:
            cout << "Unknown";
    }
}
```

6. switch with default

Output: Invalid choice

Reason: No listed case matches 4, so the default branch runs.

```
#include <iostream>
using namespace std;

int main() {
    int choice = 4;

    switch (choice) {
        case 1:
            cout << "Deposit";
            break;
        case 2:
            cout << "Withdraw";
            break;
        case 3:
            cout << "Balance";
            break;
        default:
            cout << "Invalid choice";
    }
}
```

7. switch without a break

Output: Medium Large

Reason: The 'M' case has no break, so execution falls through into the 'L' case.

```
#include <iostream>
using namespace std;

int main() {
    char size = 'M';

    switch (size) {
        case 'S':
            cout << "Small ";
            break;
        case 'M':
            cout << "Medium ";
        case 'L':
            cout << "Large ";
            break;
        default:
            cout << "Unknown";
    }
}
```

8. Multiple cases sharing one result

Output: Weekend

Reason: Cases 6 and 7 share the same output, so day 6 prints Weekend.

```
#include <iostream>
using namespace std;

int main() {
    int day = 6;

    switch (day) {
        case 1:
        case 2:
        case 3:
        case 4:
        case 5:
            cout << "Weekday";
            break;
        case 6:
        case 7:
            cout << "Weekend";
            break;
        default:
            cout << "Invalid day";
    }
}
```